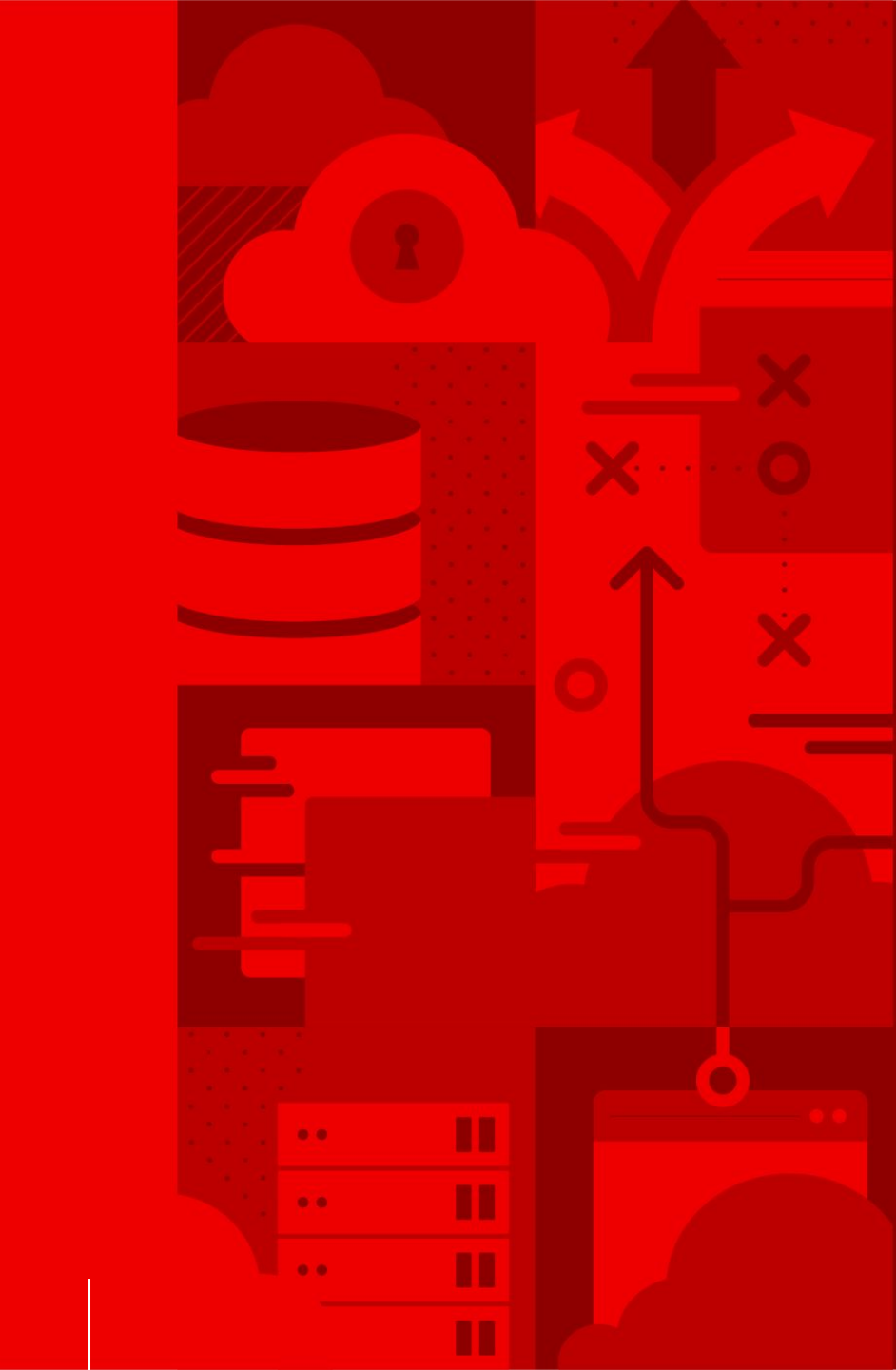




OpenShift or VMware vCloud Foundation – Virtualization

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Jonathan Gershater

Senior Principal Product Marketing Manager, OpenShift

Summary - VMware or Red Hat

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Summary

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Red Hat advantage

- Backed by IBM
- Contributions to open source demonstrating leadership and support.
- Large partner ecosystem for certified hardware and software for migrations from VMware and AI solutions.
- Published roadmap offering vision for customers.
- Security and industry certification.
- Product portfolio spanning artificial intelligence, application development, automation, containers/cloud native and Linux
- Transparent financial and headcount growth since 2000.

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Red Hat's expertise is highlighted through our leadership in open source, contribution to hundreds of open source projects, and comprehensive support and services.

Enterprise ready open source product from Red Hat	Upstream open source project
Backporting, certification, long-term lifecycles, security and 24/7 enterprise support with SLAs.	Community-driven, "best-effort" support models..

Philosophical differences between Red Hat OpenShift and VMware vCloud Foundation

- Red Hat OpenShift Virtualization 4.19 is fundamentally a Kubernetes-native platform.
 - A single, unified operational model for modern and traditional applications, managed through consistent DevOps principles, declarative APIs, and GitOps workflows, all driven by the rapid pace of open-source innovation.
 - The Kubernetes API is the universal control plane for all infrastructure resources. Built upon the open-source KubeVirt project, it treats VMs as first-class citizens within a container-centric universe, encapsulating them within standard Kubernetes objects.
- Broadcom VMware Cloud Foundation 9.0 represents the next stage of the Software-Defined Data Center (SDDC).
 - The delivery of a "modern private cloud" experience offering self-service consumption of the public cloud with the security, performance, and control of on-premises infrastructure.
 - Integrating and simplifying the management of vSphere, NSX, and vSAN—under a new, unified management framework.

Red Hat and VMware basic products

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VMware	Red Hat <i>approximate</i> equivalent
vCloud Foundation +Tanzu App Platform add-on (extra cost)	OKE + ODF + Ansible + OpenShift Virt OCP for App Services
Tanzu App Platform (minimal investment from VMware)	Developer Suite (Red Hat Developer Hub + TAS + PFA)
vSphere Foundation	OKE + OpenShift Virt

vCloud Foundation 9, what's new

VCF 9 – VMware vCloud Foundation 9

What's new?

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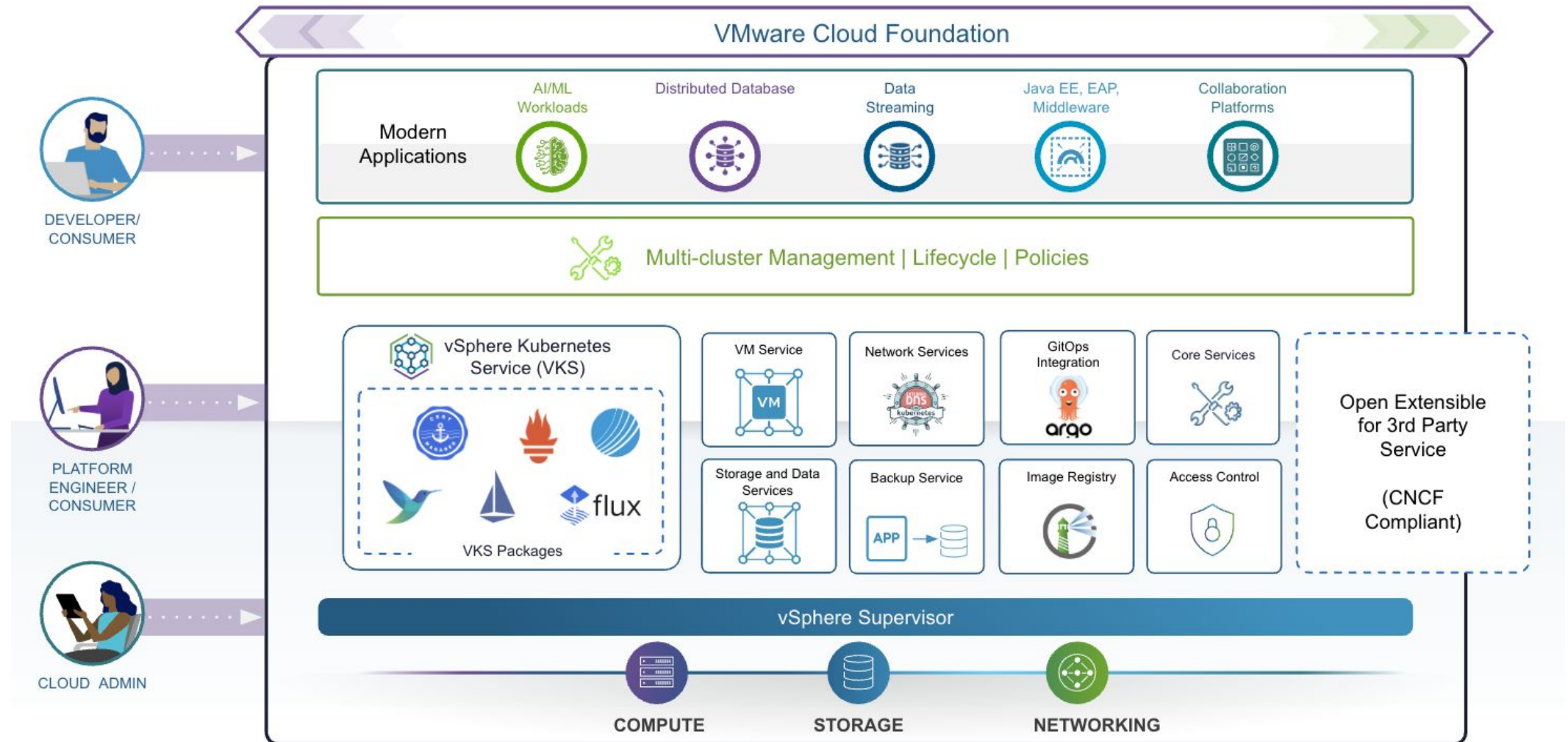
- VCF 5.2 → VCF 9
 - VCF 5.2 EOL Oct 2027
 - An **on-premise private cloud experience**. Consolidation of many disparate VMware products & UIs
- VKS
 - vSphere Kubernetes Service
 - ~~Tanzu~~ (Tanzu Application Service still exists)
- **Licensing**
 - Instead of 11 license keys, there are only two licenses for VCF – "VMware Cloud Foundation (cores)" and "VMware vSAN (TiBs)"
 - *License usage must be submitted from VCF Operations every 180 days, or hosts will disconnect from vCenter and new workloads cannot be started*
 - VCF Operations handles licence assignment.
 - If you *exceed your licensed entitlement*, additional hosts are given a 90-day evaluation licence, giving you time to either acquire more cores or decommission other hosts.
 - *(by contrast Red Hat subscriptions do not disable functionality of the software even after subscription expiration)*
- **One control plane**
 - Run VMs and Containers/K8s side-by-side.
 - Integrated Argo CD and native CI/CD *but VMware are not in the top 20 contributing organizations*
 - Istio included, *but VMware are not in the top 20 contributing organizations*
- VCF Operations – for Cloud Admins.
 - **One management plane**
 - vSphere, NSX, vSAN, Aria Automation, Aria Operations, and SDDC Manager
- VCF Automation – for private cloud Consumers.
 - **Self-service capabilities**, including provisioning databases and other resources.



VCF 9 - VMware vCloud Foundation 9

What's new?

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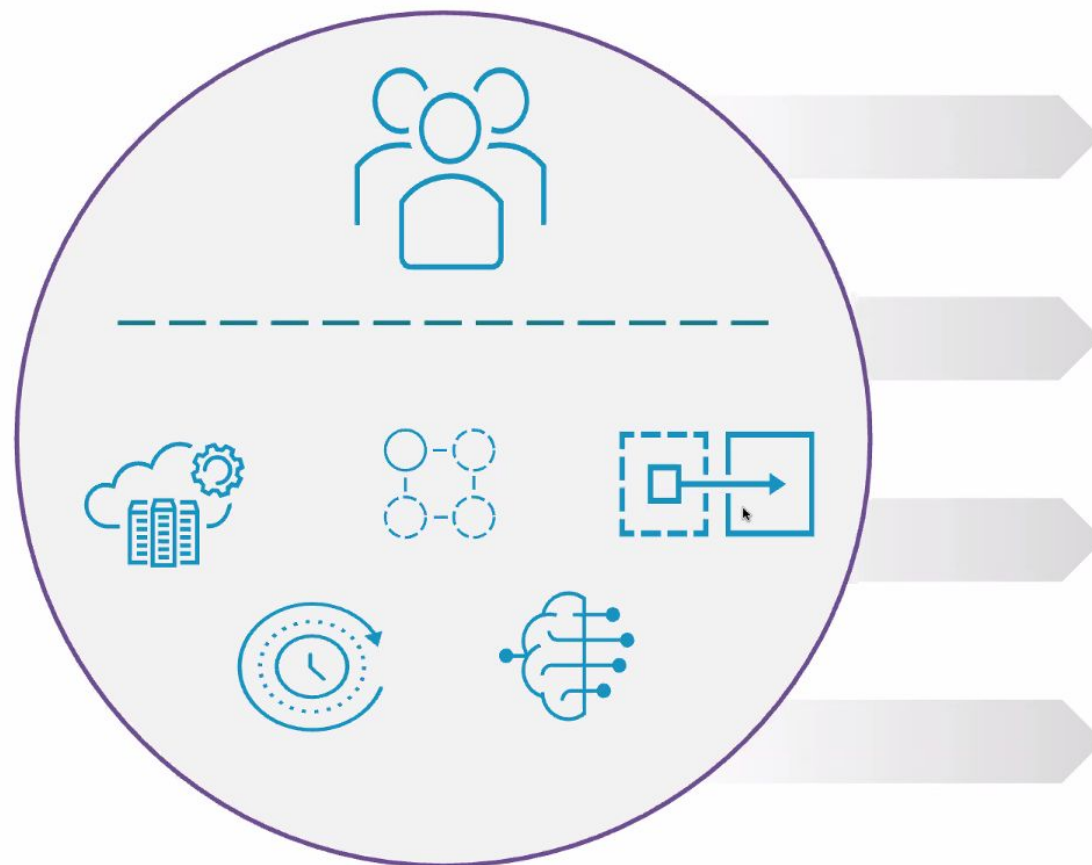


VCF 9 - VMware vCloud Foundation Operations 9

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What's new?

Operations



VCF Health & Diagnostics
(Skyline)



Integrated Log Operations
(Aria Logs)



Integrated Network Operations
(Aria Network Insight)



Integrated Storage Operations



Integrated Troubleshooting & Observability



Security Operations

Virtualization – VMware or Red Hat

- OpenShift Virtualization is **not** a rip and replace for VMware.
- The admin experience with OpenShift Virtualization will **not** mimic the VMware vCenter console.
- VMware administrators who embrace OpenShift will upskill to modern cloud-native technology.
 - ArgoCD, Kubernetes, GitOps, Istio, Tekton etc

Is OpenShift Virtualization a fit for your organization

An organization adopting OpenShift Virtualization is implicitly committing to building or expanding a team of engineers fluent in Kubernetes, who will manage VMs and containers using a unified set of tools and practices, with the goal of modernizing their IT infrastructure for the future.

An organization adopting VCF 9.0 can maintain a traditional structure, where a core team of vSphere administrators has tools to deliver Kubernetes clusters as a service to application teams.

The decision-maker is not just choosing a technology, but answering a critical strategic question: "Are we building a team of Kubernetes engineers who can also manage VMs, or a team of vSphere administrators who can also provision Kubernetes clusters?"

Partners

- Red Hat fosters a vast, open, and collaborative ecosystem. This open approach provides customers with extensive choice, allowing them to select from a wide range of certified solutions for networking, storage, security, and application services. It also ensures a broad base of community and commercial support.
- Broadcom is executing a deliberate and significant contraction of the VMware partner ecosystem, moving from a wide, tiered program to a smaller, more focused, invite-only model. This is a high-risk move that creates significant uncertainty for customers whose current service providers or resellers are not invited into the new program.

Core technical differences - Virtualization

Feature	Red Hat OpenShift Virtualization 4.19	Broadcom VMware Cloud Foundation 9.0
Core Philosophy	Kubernetes-Native Unification	Evolved SDDC Modernization
Primary Abstraction	The Pod (containing a VM)	The Virtual Machine
Control Plane	Kubernetes API Server	VCF Operations / SDDC Manager (overseeing vCenter)
Virtualization Tech	KVM/QEMU via libvirt on a Linux host (RHCOS)	ESXi Hypervisor
Unification Method	Encapsulating VMs as Kubernetes Custom Resource Definitions (CRDs)	Integrating Kubernetes services (vSphere Kubernetes Service) into a hypervisor-centric stack

OpenShift Virtualization is not a hypervisor; it is a feature of the OpenShift Container Platform that extends the Kubernetes API to manage virtual machines.

Containers (via VMware Kubernetes) are a workload type running **on** the vSphere platform, in OpenShift, VMs are a workload type running **in** the Kubernetes platform.

Why choose OpenShift Virtualization

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OpenShift Virtualization and VMware vSphere offer distinct approaches to virtualization.

OpenShift prioritizes cloud-native integration while VMware focusing on traditional virtualization. needs.

OpenShift Virtualization, leverages KubeVirt on Kubernetes to offer a unified platform for containers and virtual machines, simplifying management and promoting a cloud-native environment. VMware vSphere platform provides VM environments for VM-centric workloads.

Advantages of OpenShift Virtualization:

- Offers greater flexibility in managing hybrid workloads.
- For organizations embracing cloud-native architectures and seeking to modernize their application stack.
- Built on open-source technologies, offering greater flexibility and reducing vendor lock-in.
- Provides tools and features that simplify development workflows

vSphere products & Red Hat

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VMware Cloud Foundation (VCF)	VMware vSphere Foundation (VVF)	vSphere Enterprise Plus (EP)	VMware vSphere Standard (VVS)
Most comprehensive hybrid cloud offering. Includes vSphere, vSAN (1TiB per core included), NSX, Aria Suite (includes the original vRO, vRA, vROps, and vRealize Log Insight), HCX, DSM, TKG, and SDDC Manager. Customers using certain popular features such as NSX, vRO/vRA that used to be a-la-carte add-ons prior to 12/2023 can only get these now in VCF.	Enterprise-grade workload platform for mid-sized to smaller customers. Includes VMware vSphere, vCenter Standard, vSphere Kubernetes Service, Aria Suite Standard (Aria Operations, Operations for Logs). 11/2024 update: Price increase from \$135/core. Also, 2.5x increase in vSAN capacity included (250GB/core, up from 100GB/core).	Newly re-introduced in November 2024, VMware brought back vSphere Enterprise Plus as a standalone subscription offering. Provides the full range of vSphere features, including advanced virtualization capabilities like vMotion, DRS, VDS and high availability. Different pricing (core-based subs) vs legacy pre-12/2023 version of Enterprise Plus licenses (perpetual socket licenses, up to a 32-core limit).	Only for smaller customers using VMware for very basic hypervisor functionality. Not relevant for most enterprise customers as it doesn't have common required features like DRS, Storage DRS, and VDS.
No OpenShift equivalent. Depending on features used, partner + OpenShift with ACM+AAP+3rd party software+consulting would map to this.	Depending on what VMware features are used, OpenShift with ACM+AAP+3rd party software+consulting would map to this.	OVE would map to this, with full OCP as an upsell.	

Red Hat or VMware

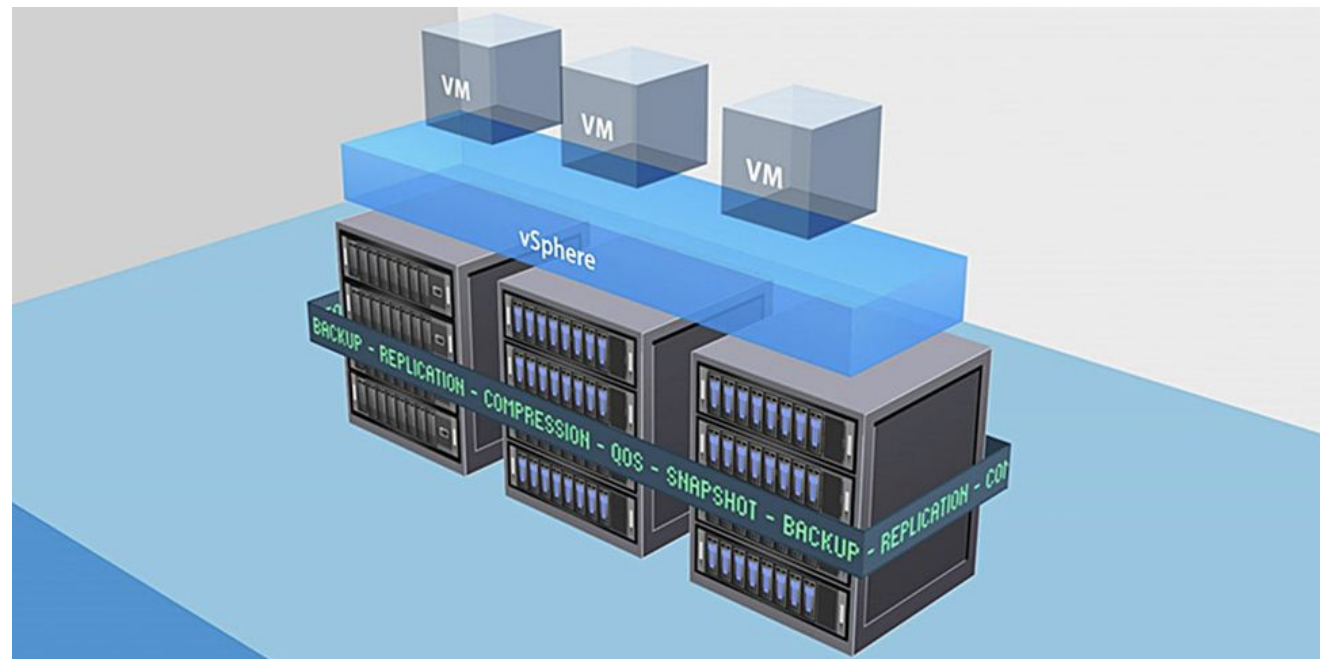
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VMware	Red Hat <i>approximate</i> equivalent
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vSphere Foundation	OKE + OpenShift Virt
vSphere	OVE

VMware vSphere Product Line Comparison

VMware is a datacenter virtualization solution

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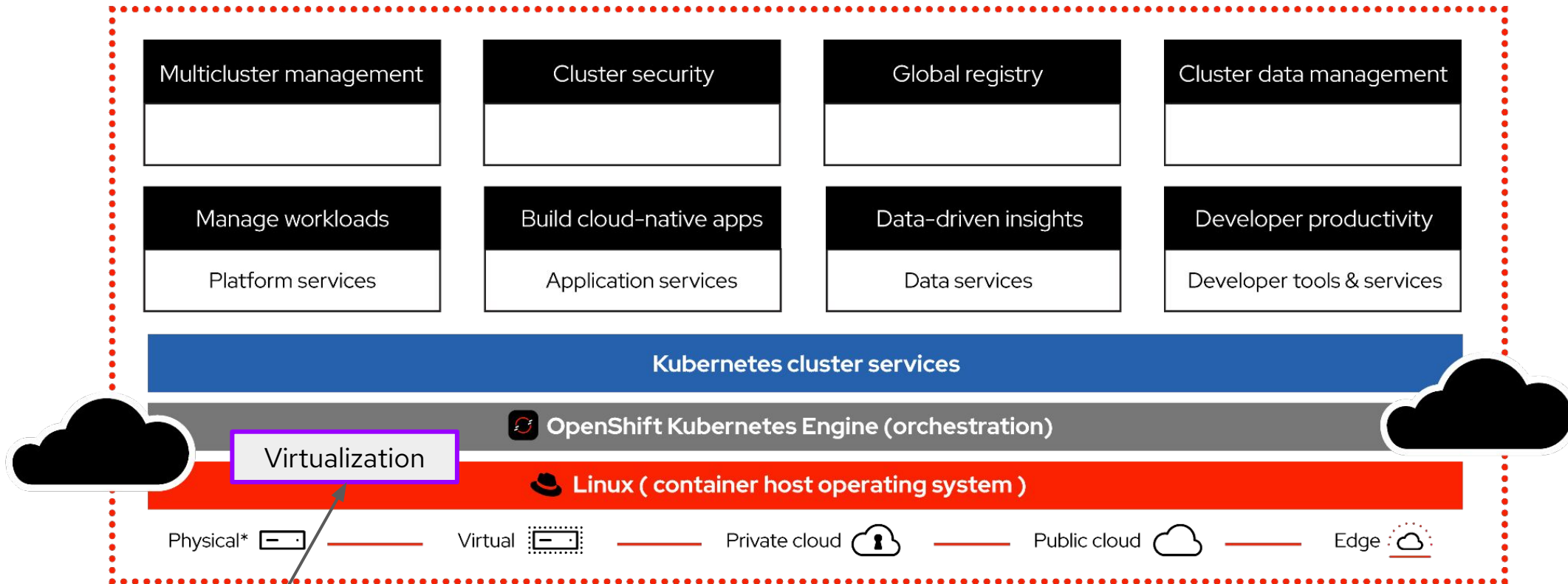


Virtualize the datacenter

- Compute
- Network
- Storage

OpenShift is an App Platform

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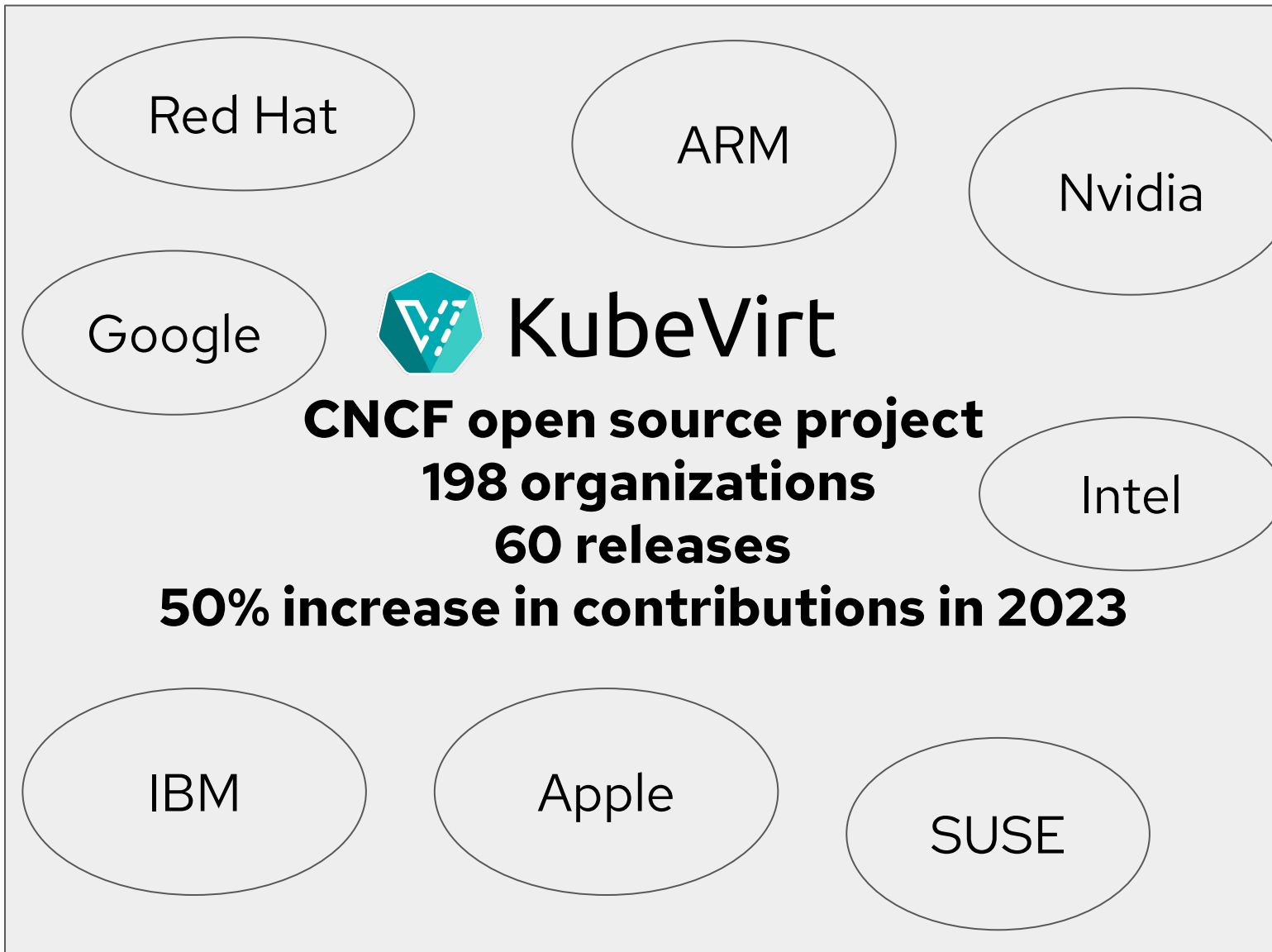


An OpenShift feature
built on KubeVirt

A modern cloud native application platform

Virtualization: Kubevirt or VMware

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NVIDIA spoke at a recent OpenShift Commons conference detailing their usage of kubevirt to run their GeForce NOW cloud gaming service. NVIDIA GeForce NOW:

- 28 Million users
- 1500+ Games
- 100+ Countries
- 30+ Data Centers



Job trends, open source or VMware



Jobs ▾ | **Date posted** ▾ | **Experience level** ▾

open source in United States Set
8,824 results



Jobs ▾ | **Date posted** ▾ | **Experience level** ▾

vmware in United States Set a
657 results



Jobs ▾ | **Date posted** ▾ | **Experience level** ▾

kubernetes in United States Set
25,266 results



Jobs ▾ | **Date posted** ▾ | **Experience level** ▾

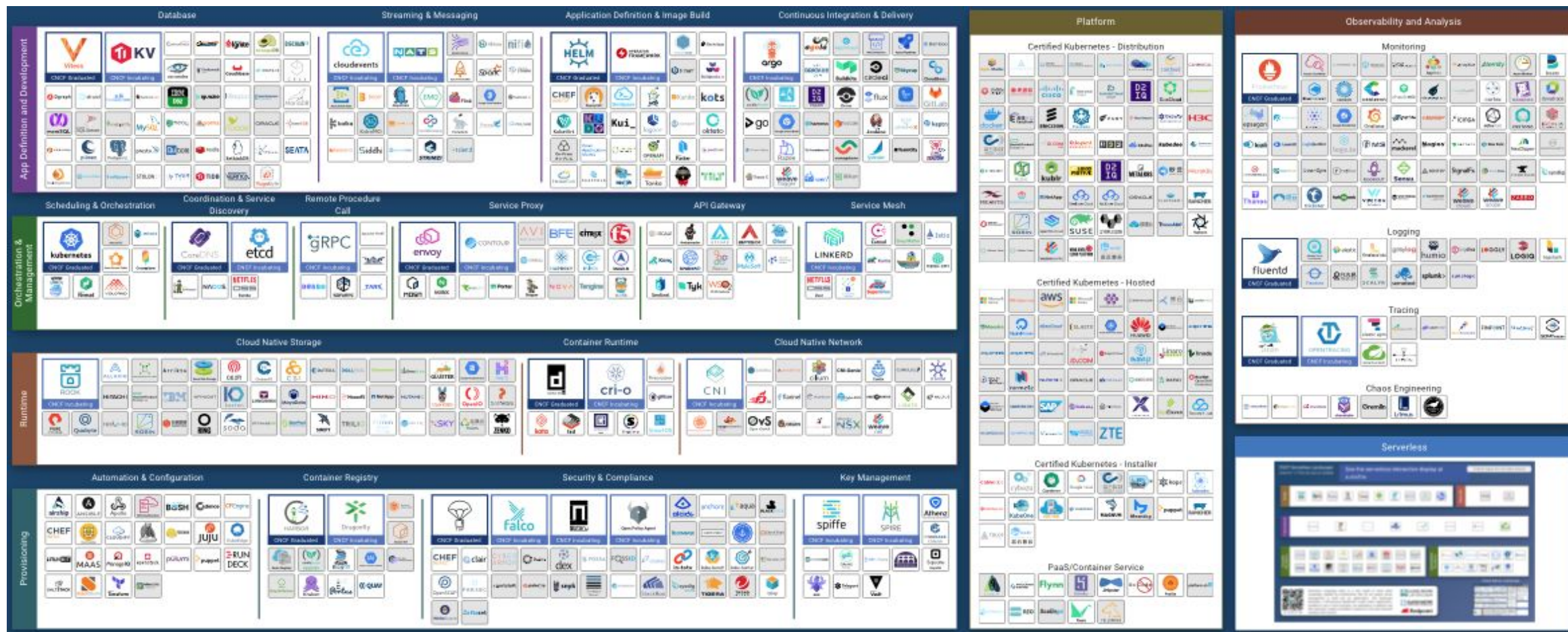
vsphere in United States Set
8,401 results

21

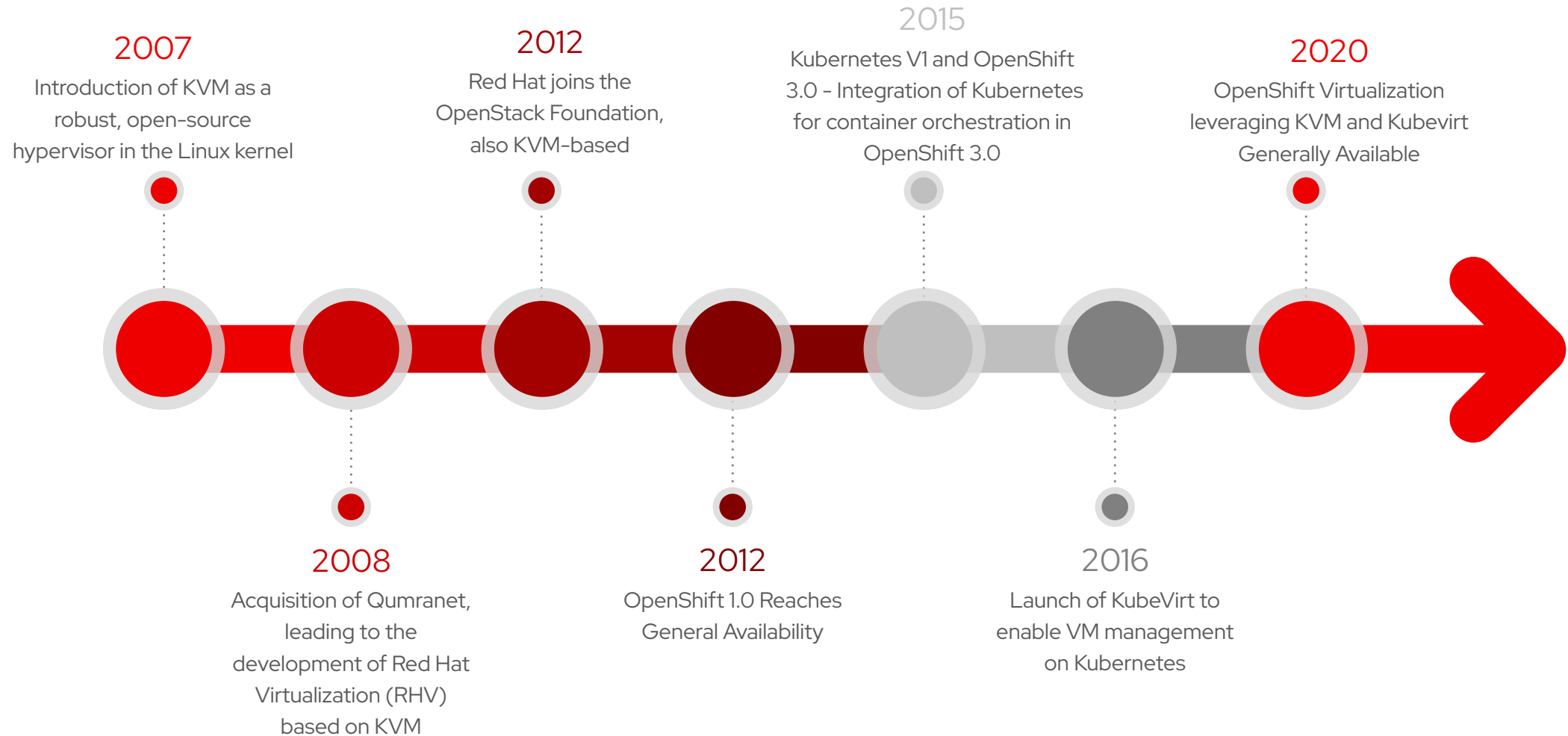
Jobs are trending from VMware/vSphere to Kubernetes / open source (LinkedIn snapshot, November 2023)



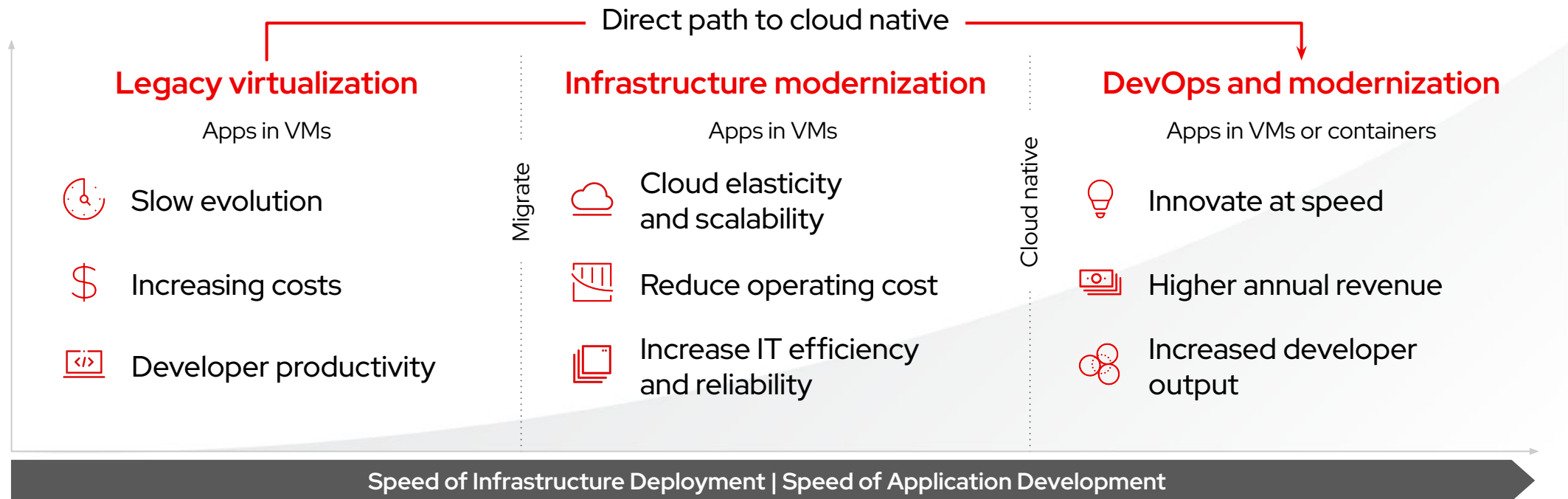
Cloud Native Computing Foundation



Red Hat has a long history with Virtualization



Modernize at your own pace



Goldman Sachs & Kubevirt

Goldman Sachs discusses virtues of Kubernetes for virtual machine management

Kubernetes is not only a platform for orchestrating containers – Goldman Sachs has found it offers the same benefits when used to manage virtual machines



By **Cliff Saran**, Managing Editor

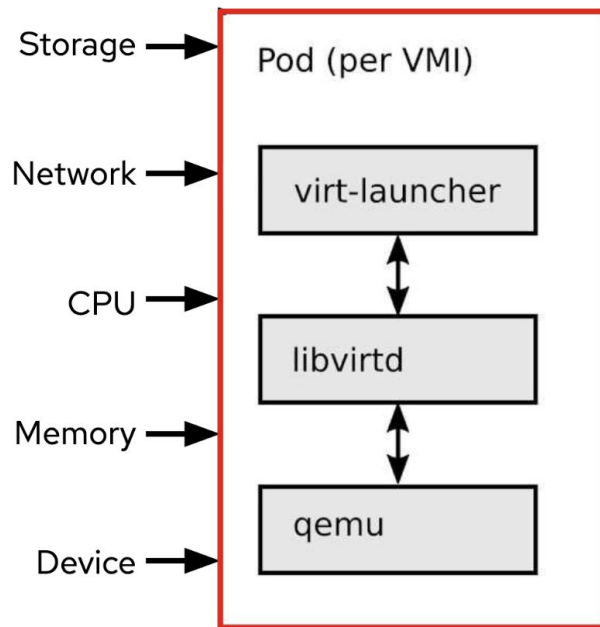
Published: 26 Mar 2024 12:22



Red Hat OpenShift virtualization, technical

Modernize infrastructure and support mixed applications consisting of VMs, containers, and serverless

Containerized virtual machines



Kubernetes resources

- Every VM runs in a launcher pod. The launcher process will supervise, using libvirt, and provide pod integration.

Red Hat Enterprise Linux

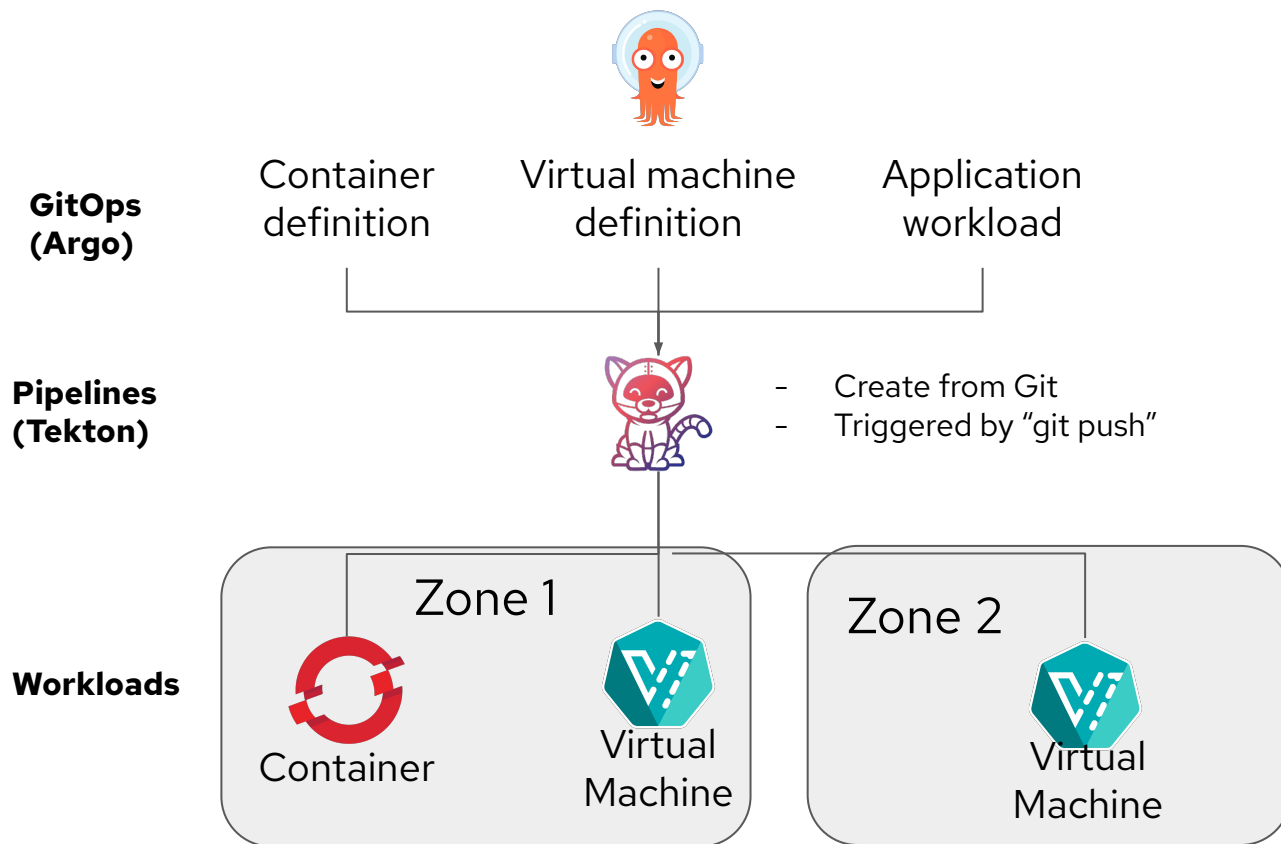
- libvirt and qemu from RHEL are mature, have high performance, provide stable abstractions, and have a minimal overhead.

Security - Defense in depth

- Immutable RHCOS by default, SELinux MCS, plus KVM isolation - inherited from the Red Hat Portfolio stack

OpenShift Virtualization: Build Cloud-native VMs

Deploy VMs as Code with CI/CD



Integrate legacy VMs with a modern GitOps framework

- ▶ Deploy different security zones to run both composite applications of pods/VMs as well as traditional VM workloads
- ▶ Deploy and automate Virtual Machines as Code with GitOps

Red Hat OpenShift virtualization on AWS (ROSA)

Modernize infrastructure and support mixed applications consisting of VMs, containers, and serverless

- OpenShift Virtualization on ROSA offers mixed environments of Linux, Windows VM's, container or serverless workloads in the same name space, network domain and automation framework
- Apply CI/CD driven DevOps/GitOps workflows to your virtual machine workloads in exactly the same model you enjoy with cloud native containers today. Tekton, ArgoCD, Scupper, Mesh Network, Dev Spaces, Secrets, Container Registry, Knative and other open source tools pre-integrated into OpenShift treat infrastructure as code for traditional App/VM workloads
- VM features like Live-Migration and cross pod access, use RWX capable native AWS storage such as the AWS FSX NFS service or one of our partner storage solutions.
- OpenShift Virtualization is storage agnostic and can integrate with multiple storage partners; expect future blogs from these partners . Data Protection is provided with local copy persistent storage as well as with partnerships for Data Protection that support OpenShift Virtualization. These partnerships can facilitate migration to ROSA with their associated replication capabilities from their data center solutions
- OpenShift Virtualization on ROSA fully supports Linux and Windows images with 3rd party support for other OS options. A key use case is the ability to migrate Windows .NET applications on Windows to ROSA and then rapidly modernize them in pieces to containers with scale-out and more reliable options. Other options such as modernizing Spring Boot apps, Herokou are other examples for conversion but migrating any app to a container architecture will show benefits in the cloud on cost, flexibility and scaling. Many commercial off-the shelf applications also enable container modernization on OpenShift such as IBM Maximo, IBM MQ, Mulesoft to name a few

Windows Virtual Machine Modernization on ROSA

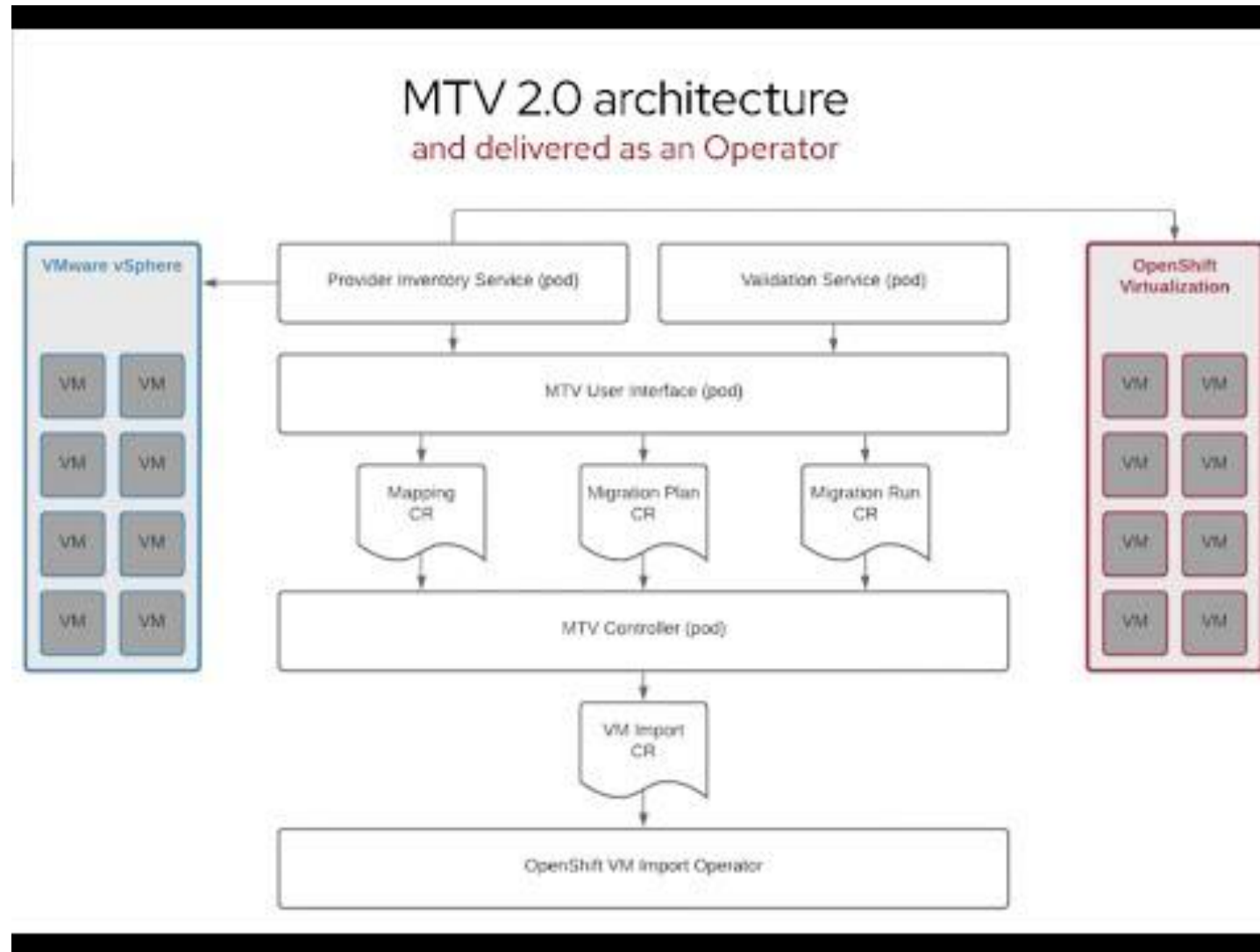
The challenge of containerizing Windows VM on AWS EKS

- Windows containers are not available on AWS/EKS
- EKS requires Windows Workers with limited supportability
- Windows workloads typically must run on EC2 instances

With OpenShift Virtualization on ROSA

- Run and Manage Windows VMs side by side to containers
- Use the same network and storage as your containers
- Deploy using the same tooling as containers
- No need to rewrite the application
- Modernize portions of the application

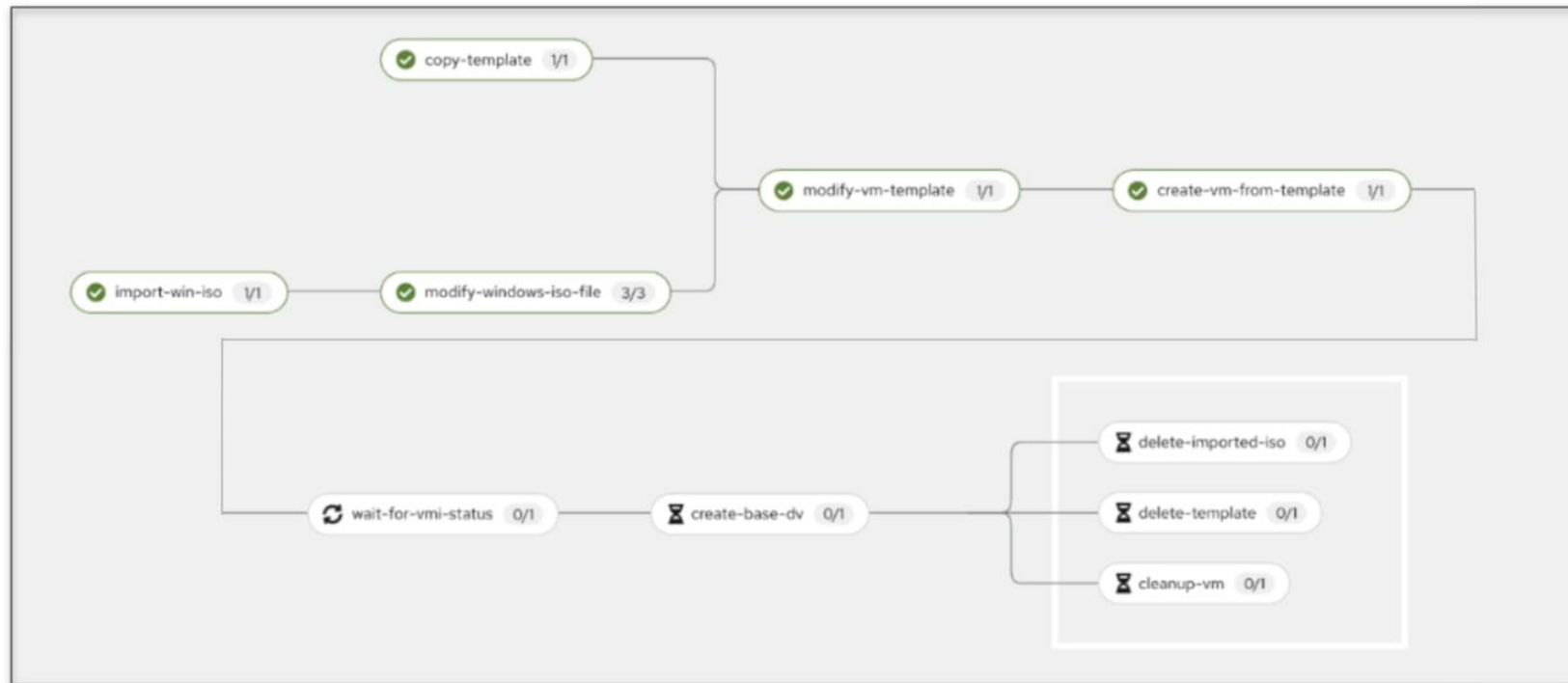
Migrating Virtual Machines to OpenShift Virtualization



Managing VMs using cloud native technology for AI/ML

Using OpenShift Pipelines

- OpenShift pipelines can control many VM tasks, from a single VM operation to managing the VM's entire life cycle: Downloading a Windows OS image, customizing VM configuration, deploying the VM, and more. This creates VMs that are consistent and that minimize human errors and increase security.
- AI/ML workloads may require VMs for AI/ML visualization. These VMs can be fully provisioned and managed by OpenShift pipelines, from image preparation, attaching a data source, starting the VM that runs the ML module, and once done, shutting down the VM (and in some cases, deleting it). Repeat this pipeline with a new data source as many times as needed.



<https://cloud.redhat.com/blog/running-vms-the-cloud-native-way>



Managing VMs on OpenShift in AWS

Using OpenShift Pipelines

Support Legacy Applications

- Many mission-critical applications run on older operating systems that are not supported on cloud platforms and cannot be easily containerized. It's impractical to rewrite and modernize all applications into containers.
- These VMs on OpenShift, means the applications can be integrated with containers
- Customers can take advantage of the benefits of cloud-native architectures without sacrificing the functionality of their VMs.

Lift-and-Shift Migration

- With OpenShift Virtualization, customers can take advantage of a swift lift-and-shift method to move to a consolidated platform without the need for significant code refactoring, Including Microsoft Windows VMs.
- If the VM operating system is not supported natively on AWS EC2, it is support on OpenShift

Accelerate Application Delivery with a Single Platform

- Developers can create VMs for testing and development, reducing the time and cost of the development process, with a unified DevOps pipeline.
- OpenShift provides a set of unified developer capabilities across VMs and containers, reducing the tools and runtimes developers need to use to successfully build and deploy enterprise-grade software.

In-Place Modernization

- Wrap VM legacy applications into a container pod, deploy on OpenShift, and gradually modernize application services.
- Implement CI/CD pipelines, service mesh, monitoring and logging, scalability, resiliency, and continuous improvement.



VMware Red Hat mapping

- This is **not** a feature comparison
- OpenShift Virtualization does **not** map features 1:1 with VMware
- Red Hat offers alternative solutions, often using modern cloud native methodology.

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Please see [this doc on RHCC](#) for a mapping of OpenShift Virtualization and VMware

Thank you

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